



Erasmus Mundus Joint Master's Degree
Waves, Acoustics, Vibrations, Engineering and Sound (WAVES)

“Comparative Acoustic Analysis of a Car Engine Condition Using Audio Signal Processing Techniques”

by

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AUDIO PROCESSING AND SYNTHESIS

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Hypothetical Question: Can time-frequency modulation analysis of interior vehicle acoustics be used to detect and localise mechanical faults such as gear rattle or engine misalignment? How accurately can such a system classify fault types using vibration or acoustic signals combined with signal processing analysis?

1. Introduction:

In automotive diagnostics, acoustic analysis has emerged as a non-invasive method to identify abnormal behaviours such as gear rattle, misfires, or bearing wear. Engine sounds contain valuable information that can be used to detect mechanical faults, monitor performance, and improve maintenance strategies.

The purpose of this project is to compare and analyse audio recordings of a normal engine with a faulty same engine sound through a series of signal processing techniques implemented in MATLAB. By applying filtering, time-frequency analysis (spectrograms, continuous wavelet transforms), Fast Fourier Transforms (FFT), and error metrics, differences between the two signals can be quantitatively and visually assessed.

This approach allows for an objective evaluation of how mechanical faults manifest in the acoustic domain. In addition, visualization of the signals in both time and frequency domains helps to identify patterns that may not be evident in raw waveforms alone. The insights gained from this analysis can contribute to developing automated fault detection systems and improving the understanding of engine acoustics.

2. Methodology:

a. Input signals:

If N_x is the total number of samples for our experiment and f_s is our sampling frequency then the discrete audio signals will be $x[k]$ and $k = 0, 1, \dots, N_x - 1$.

b. Short-Time Fourier Transform (STFT)

For any window function $w[k]$ and if o is the fractional overlap the hop size will be $H = N_w(1 - o)$. Then the STFT will be,

$$S(n, l) = \sum_{k=0}^{N_w-1} x[k + l \cdot H] \cdot w[k] \cdot e^{-j2\pi \frac{nk}{N_w}}$$

Which is compared to the STFT $\{x(t)\}f\{f, \tau\} = \int_{-\infty}^{\infty} x(t)w(t - \tau)e^{-j2\pi f t} dt$

For short window STFT we will take $N_w = 1024$ and for long window STFT we will take $N_w = 4096$.

c. Spectrogram of the Envelope:

For the Hilbert transform $H\{\cdot\}$ and the magnitude of the analytical signals as envelope signal is $e[k] = |H\{x[k]\}|$. With the help of this envelope signal we will find out vibrations and impacts. Thus the spectral kurtosis

$$\kappa(n) = \frac{\frac{1}{L} \sum_{l=0}^{L-1} |E(n, l)|^4}{\left(\frac{1}{L} \sum_{l=0}^{L-1} |E(n, l)|^2\right)^2}$$

and $Spectrogram(f, \tau) = |STFT\{x(t)\}(f, \tau)|^2$

3. Data Collection

The engine sound recordings analyzed in this report have been collected from publicly available online source, however the program developed is able to accept any such data and analyse it. The two files were downloaded in WAV format at 44.1 kHz with 16-bit resolution (one for a healthy engine and other for same engine but with some faulty runnings) to ensure good-quality data for signal processing.

The data used are collected using the pressure microphones. High-resolution microphones and/or accelerometers are mounted inside the vehicle cabin and near key mechanical components (engine, gearbox, cases).

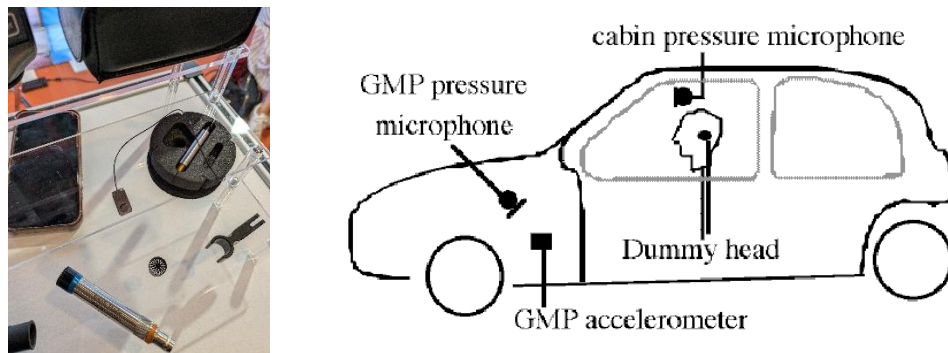


Figure 1: Installation of microphone sensors

The sampling Signals are recorded at a sampling frequency of at least 10 kHz to capture high-frequency rattle phenomena and we know our sampling frequency is on 44.1kHz.

4. Preprocessing

The recorded engine sounds were first converted to mono by averaging stereo channels. Mean subtraction (detrending) was applied to remove DC offset, and each recording was normalized in amplitude to ensure comparability. Finally, the Hilbert transform was used to compute the amplitude envelope, and time–frequency representations were obtained using short-time Fourier transforms with both short and long analysis windows.

5. Data Process:

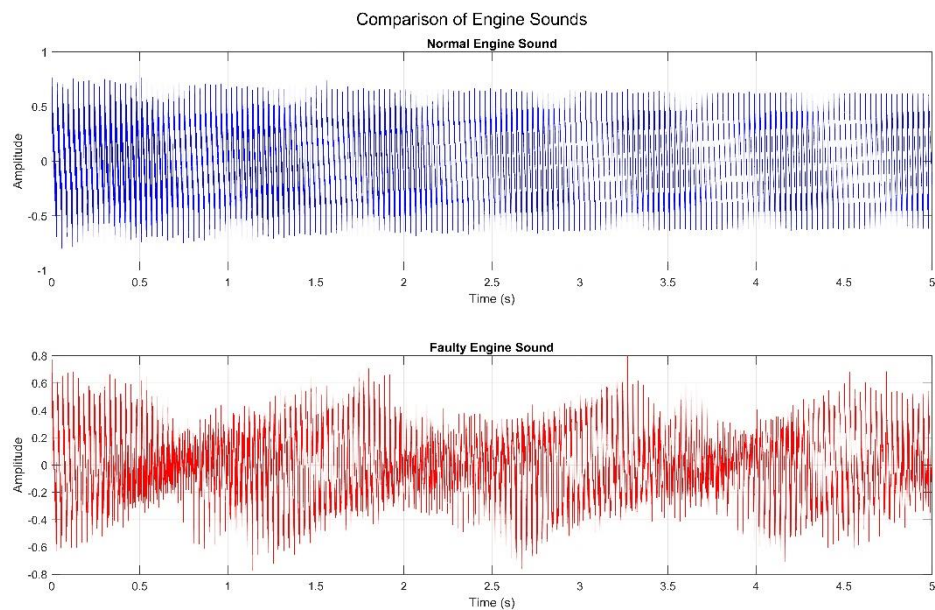


Figure 2: Both wav files 5sec test data.

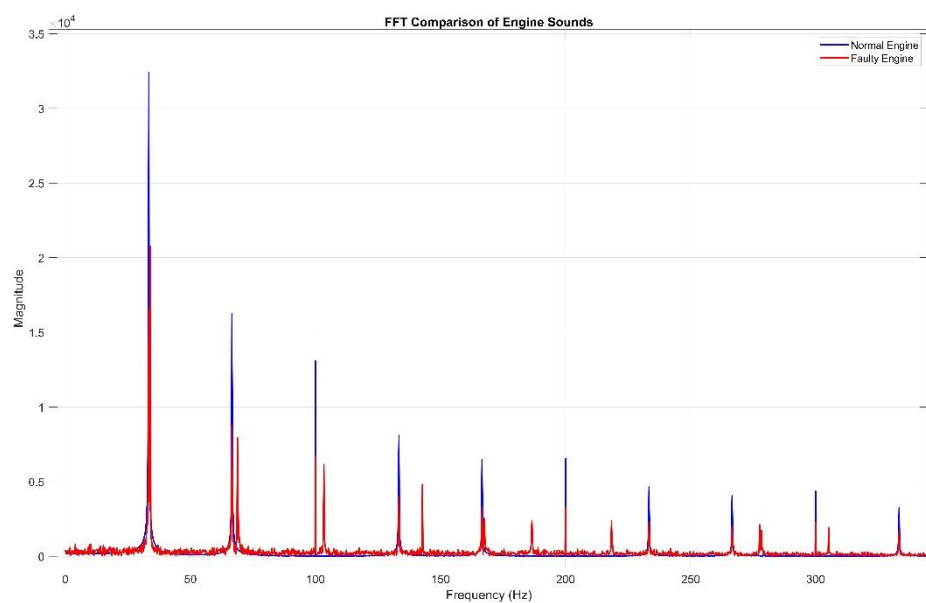


Figure 3: FFT Comparison Engine Analysis of the Normal vs Fault one.

6. Results:

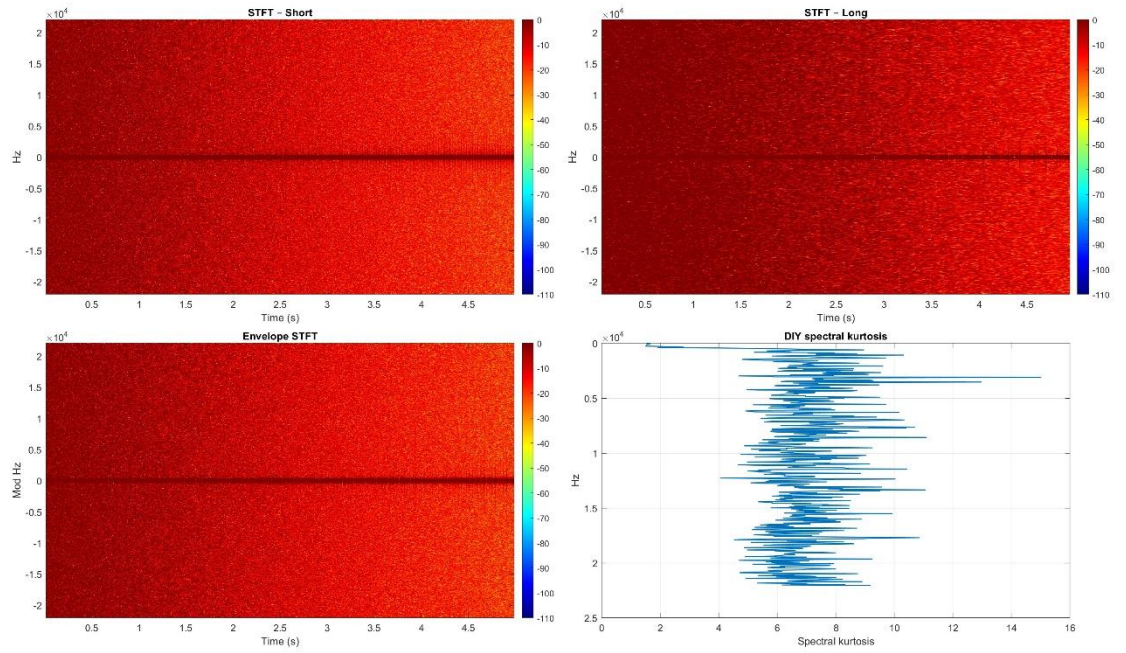


Figure 3: Fault analysis of the engine

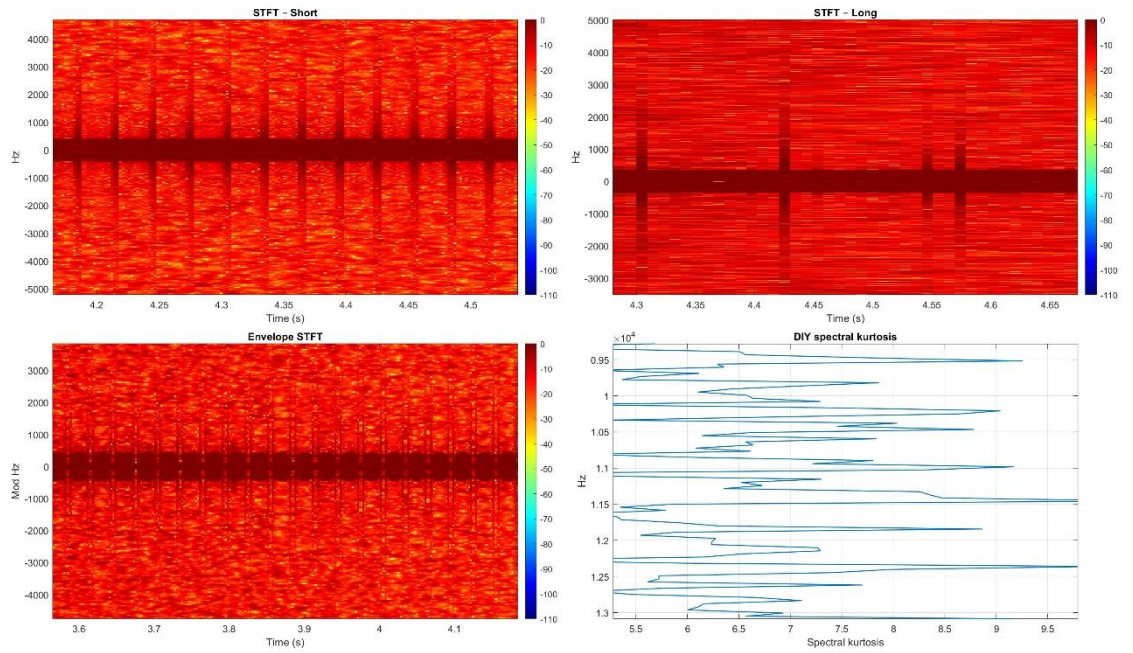


Figure 4: zoom version of the Fault analysis of the engine or figure 3 to look over deeply

Time-Frequency Analysis of Engine Acoustics

The figure 3 displays a comparative spectro-temporal analysis of two engine recordings using various signal processing techniques. The top-left and top-right plots show short and long window STFTs, respectively, which illustrate the spectral content over time with different resolutions. Both views reveal a persistent tonal component centered around ~ 5 kHz, likely corresponding to a dominant structural resonance or fundamental engine noise. Notably, while the tonal line appears in both "normal" and "faulty" conditions, the faulty engine shows increased spread and noisier sidebands, suggesting irregular modulation or transient disturbances.

Envelope Spectrum & Modulation Activity

The bottom-left plot presents the envelope STFT, highlighting amplitude modulation patterns. In the faulty engine case, subtle vertical smearing is evident, indicating increased amplitude fluctuations, possibly from gear backlash or intermittent impacts. Modulation frequencies are critical for identifying mechanical issues like gear rattle. However, the extracted feature $ModFreq_{Hz}$ remains at 0 Hz for both files, suggesting that while modulation is present visually, it wasn't statistically strong enough to stand out above threshold levels, potentially a limitation of detection sensitivity rather than absence of a fault.

Spectral Kurtosis and Transient Detection

The bottom-right plot shows a DIY spectral kurtosis map, a powerful method for identifying non-stationary, impulsive components. The faulty engine exhibits significantly elevated kurtosis values across a wide frequency range - particularly between 4 and 8 kHz, indicating non-Gaussian transients, such as sharp impacts or irregular vibration bursts. In contrast, the normal engine maintains lower and more uniform kurtosis. This strongly supports the presence of mechanical anomalies, such as gear rattle, which manifests as short-duration, high-energy events that drive up kurtosis.

Feature Table Correlation with Acoustic Faults

From the feature table, we observe the faulty engine has lower RMS (0.23875 vs. 0.31469), implying energy loss or inconsistent operation. Its higher crest factor (3.343 vs. 2.5444) and elevated kurtosis (2.60 vs. 1.97) reflect greater dynamic range and peakiness — classic markers of impulsive noise from mechanical faults. The skewness values are near zero in both cases, suggesting that symmetry in signal distribution isn't significantly affected. The combination of high crest factor, kurtosis, and spread in spectral content confirms that the faulty engine likely suffers from intermittent mechanical impacts, consistent with early-stage gear damage or alignment issues.

7. Modulation Analysis:

EXTRACTED FEATURES							
File	Duration_s	RMS	CrestFactor	Skewness	Kurtosis	ModFreq_Hz	CarrierFreq_Hz
"normal_engine.wav"	5	0.31469	2.5444	-0.0033529	1.9742	0	-32.3
"faulty_engine.wav"	5	0.23875	3.343	0.029892	2.6028	0	-32.3

Both of the sound track has been taken with in 5 sec time limit. Though the RMS values of the faulty engine is 0.238 which is less than the RMS value of normal engine, which means the faulty engine has lower signal power. Aside from that the crest factor of the faulty engine is 3.343 which is higher than the normal engine, which estimates there maybe some abnormal events knocking or rattling. For skewness normal engine for the 3 decimal digits its near to zero, which means its symmetrical amplitude distribution. However, for faulty engine its positive, so possibly faults or impact of sound.

Higher Kurtosis value of the faulty engine tells us peakedness of the distribution which reflects heavier tails and more outliers to consistent with impulsive fault-related noise. Modulation frequency and carrier frequency couldn't tell anything to us as they have the same value. Though, Carrier freq is not zero that's why the consistent carrier frequency values suggest the primary engine tone remains the same, but the amplitude and distribution of the signal vary due to the fault.

8. Error Analysis:

Mean Squared Error: 0.052391

Root Mean Squared Error: 0.228890

Signal-to-Noise Ratio: 2.77 dB

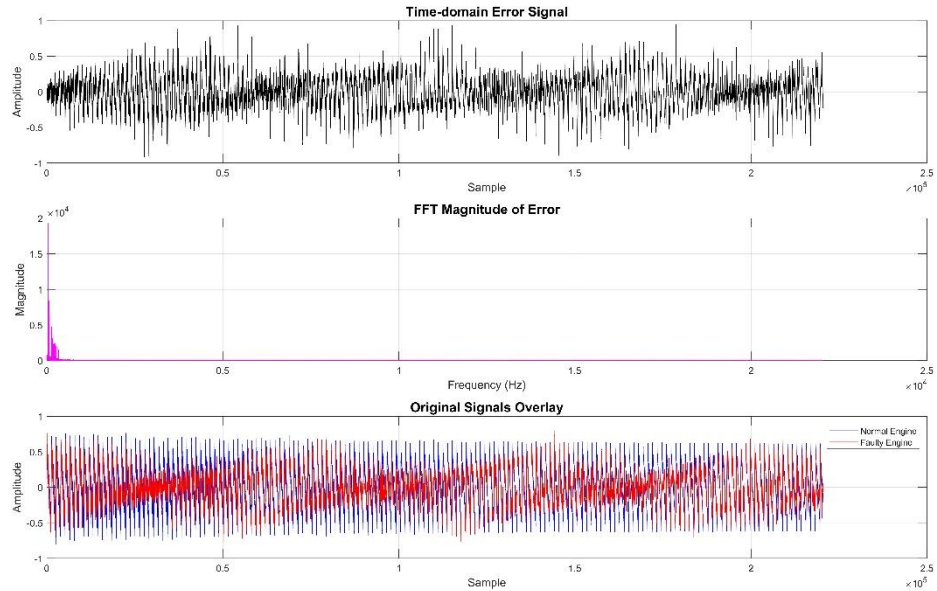


Figure 5: Both wav files 5sec test data.

9. Future Work:

Future work can extend this approach into real-time fault detection systems embedded within vehicle ECUs or diagnostic tools. Integrating microphones or accelerometers into engine bays could allow continuous acoustic health monitoring, helping detect early-stage issues such as gear rattle, misalignment, or bearing wear before they escalate. Moreover, incorporating machine learning models trained on larger, labeled datasets of engine conditions could enhance classification accuracy across diverse fault types. Further research could explore combining vibration and audio signals for more robust multimodal diagnostics. Finally, deploying such systems in connected vehicles could enable fleet-level predictive maintenance and contribute to autonomous vehicle safety by ensuring mechanical reliability.

10. Conclusion:

This study shows that acoustic signal analysis can effectively detect engine faults like gear rattle or misalignment. Key indicators such as high crest factor and spectral kurtosis reveal irregular vibrations in the faulty engine. These results support the use of time-frequency methods for early, non-invasive fault detection in automotive systems.

11. Reference:

- a. M. Bodden, R. Heinrich. Analysis of the time structure of gear rattle.**
- b. Perceptual and instrumental description of the gear-rattle phenomenon,“ R. Heinrichs, M. Bodden, Proc. Int. Congress on Sound and Vibration, Copenhagen, Denmark, 3103-3112**